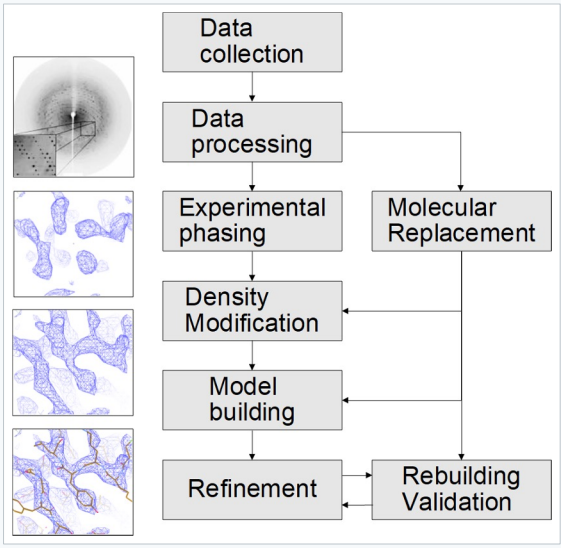
CCP4 SHANGHAI WORKSHOP

Automated Model Building

Jordan Dialpuri, PhD – MRC Laboratory of Molecular Biology

15th September 2026

Caption for this presentation were written mostly by Paul Bond, PhD – University of York. Many thanks to Paul for sharing most of the images for these slides.



Flowchart of structure solution process in X-ray crystallography.

IMAGE FROM KATHRYN COWTAN

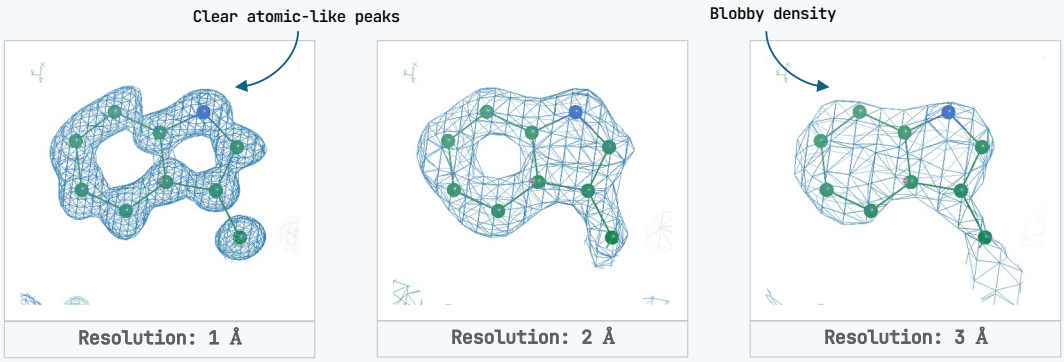
Structure solution in X-ray crystallography

Exact path depends on target under investigation

This is a rough depiction of the structure solution process for X-ray crystallography. Once you've done the hard work of growing a crystal, you need to collect images and process them to get structure factor amplitudes. In order to get an electron density map you also need to get some phase estimates either through experimental phasing or molecular replacement. If you've used experimental phasing you will need to do density modification to improve the phases and hence the quality of the map. You then use automated model building to build an initial model into the improved map, followed by multiple rounds of refinement, rebuilding and validation. However, it's more likely that you'll get phases from molecular replacement (MR). If your MR model is poor then you might need to do density modification first before you can improve it with automated building. If the MR model is very good then you might even want to skip automated building and go straight to interactive rebuilding and validation. Automated model-building pipelines try to automate the steps in the red box as much as possible.

IMAGES FROM KATHRYN COWTAN, SOON WEN HOH AND PAUL BOND

Appearance of electron density with varying resolutions



Appearance of electron density with 1 to 3 Å resolution.

A big problem with automated model building is that electron density maps look very different at different resolutions. At high resolution you can distinguish peaks for individual atoms, but if you write a program that looks for separate atomic peaks then it wont work when you have a low resolution map and you can't see them. The local resolution gives an indication of the map quality in that region. Tryptophan has a large, rigid side chain that makes it easier to spot in the density. At lower resolutions, most side chains are not identifiable without previous knowledge of the sequence.

ALL IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP

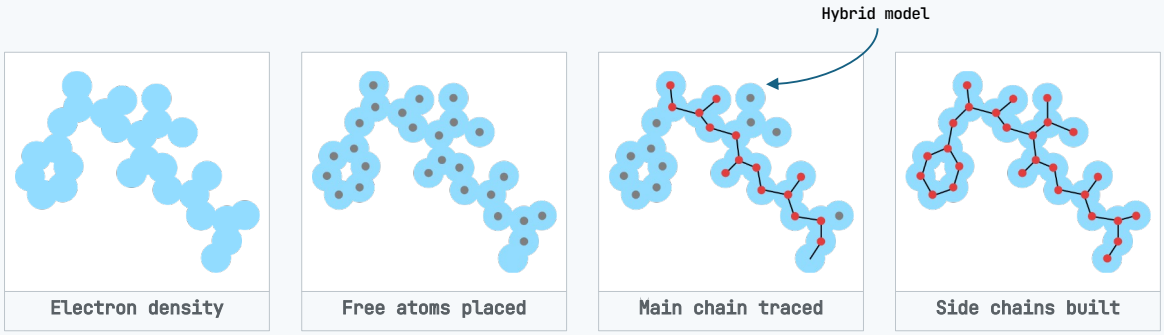
Victor Lamzin - EMBL Hamburg
Grzegorz Chojnowski - EMBL Hamburg
Anastassis Perrakis - Netherlands Cancer Institute



The first program we will look at is ARP/wARP, which was the first program to provide fully-automated model building from an empty map to a complete model. Thank you to Grzegorz Chojnowski who provided these slides in the presentation.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP uses a free-atom model to locate amino acids

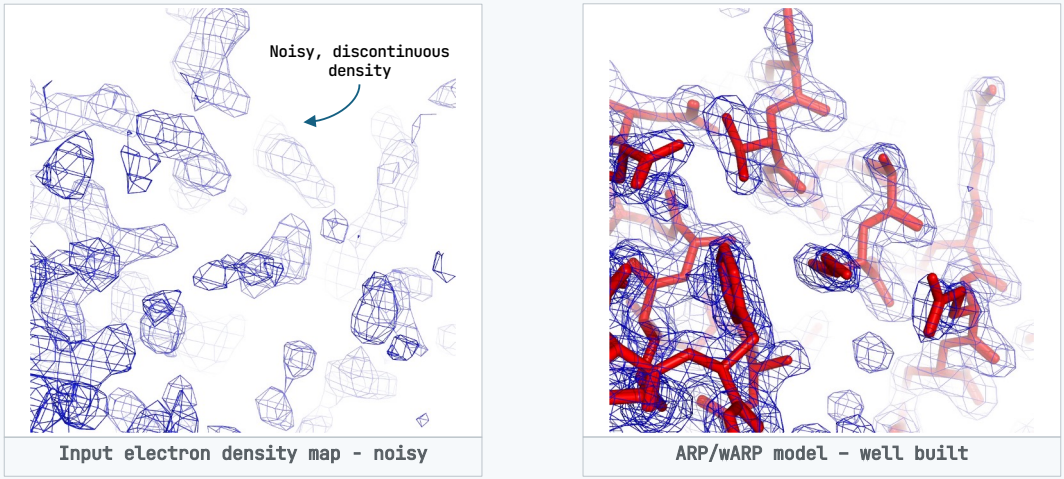


Schematic of free-atom placement, main-chain tracing, and side-chain tracing in ARP/wARP.

The first step in ARP/wARP is to build and refine a free-atom model into the map (where the atoms do not have any chemical identity). The main-chain is then traced through the free atoms, then the chains are sequenced and side chains are built. The process is repeated in cycles so you will have a mixed model that contains both built residues and free atoms. Once the chains have been assigned to the sequence, missing regions are identified and built.

IMAGES FROM GREZEGORZ CHOJNOWSKI

An example of a successful ARP/wARP run

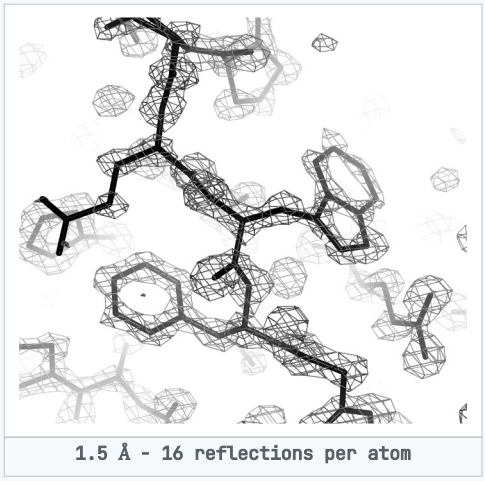


Automated model building of a protein with ARP/wARP.

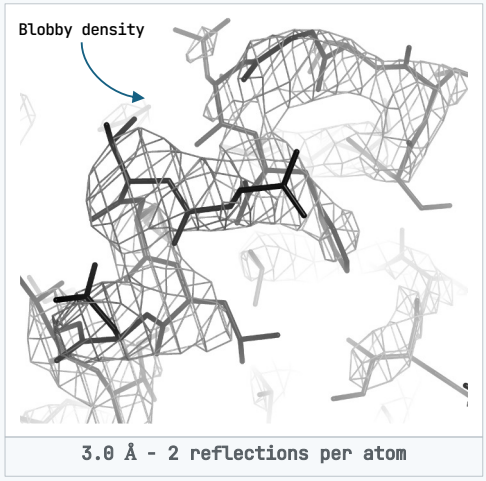
Here is a real example at 2.0 Å resolution showing the input and output of ARP/wARP. It has gradually added free atoms to the map, improved their positions and interpreted them as an atomic model of a protein. You can see by comparing the maps on the left and right that the quality of the map itself also gradually improves throughout this process as the new model provides new phases (in X-ray crystallography).

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP has a large resolution dependence



1.5 Å - 16 reflections per atom



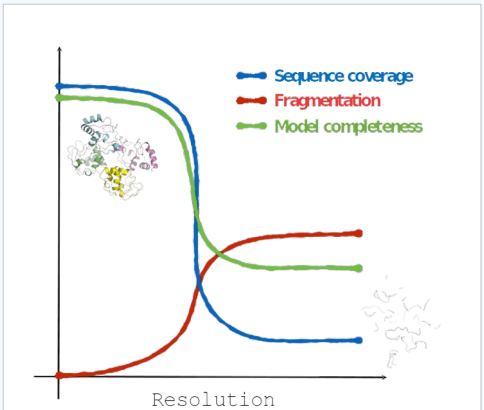
3.0 Å - 2 reflections per atom

Density appears more clear at higher resolutions, allowing ARP/wARP to work well.

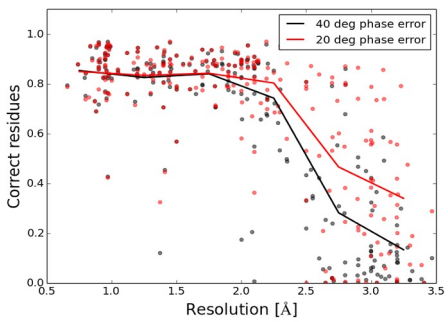
The problem with this approach is that at low resolution there is not a much information about individual atoms. At 1.5 Å resolution there might be ~16 reflections per atom but at 3.0 Å resolution there are probably only ~2 reflections per atom. This means that each atom will not have it's own peak in the map and so getting accurate free-atom positions is less likely.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP has a large resolution dependence



Performance of ARP/wARP with resolution



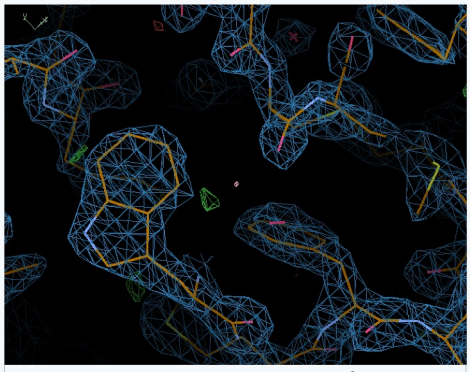
Performance of ARP/wARP with resolution

ARP/wARP performance drops after 2.5 Å resolution

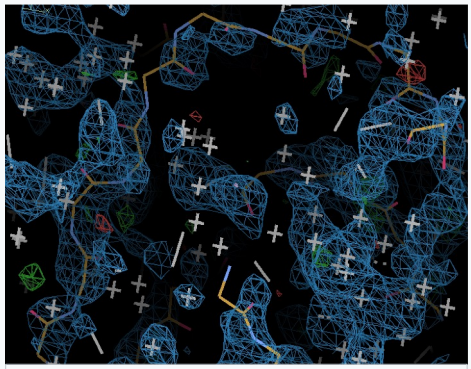
At low resolution the model becomes less complete, more fragmented and it is more difficult to assign the correct sequence to the parts that have been built. The chart on the right shows some real examples at different resolutions. With resolutions better than 2.0 to 2.5 Å, classic ARP/wARP is usually able to build a very complete model even if the phases are initially quite poor. It is still possible to build some of the model, but there is more chance of this with more accurate phases.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP has a large resolution dependence



Successful model from ARP/wARP



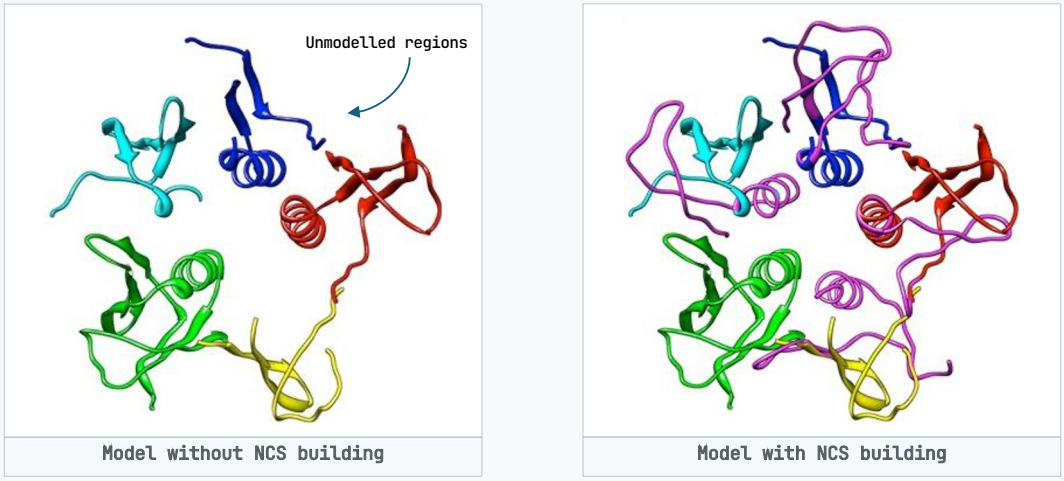
Unsuccessful model from ARP/wARP

ARP/wARP may output a very nice model or the remnants of the hybrid model.

The model on the left shows a good model from ARP/wARP. If the model reaches a certain quality the free atoms are removed from the output structure. On the right is an example of a poor model from ARP/wARP. The model is very fragmented and there are also a lot of unidentified free atoms, although some of the structure may still be correct.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP can detect and extend NCS-related chains



If ARP/wARP performs badly, recovering sections using NCS building could be helpful.

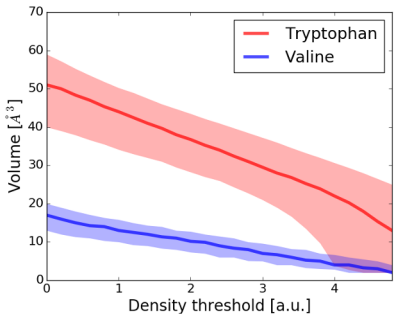
There have been a number of improvements to the classic ARP/wARP method. One is able to detect that partially-built chains are non-crystallographic (NCS) copies of each other and use the better copies to help complete the worse ones.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP can assign sequences without free atoms



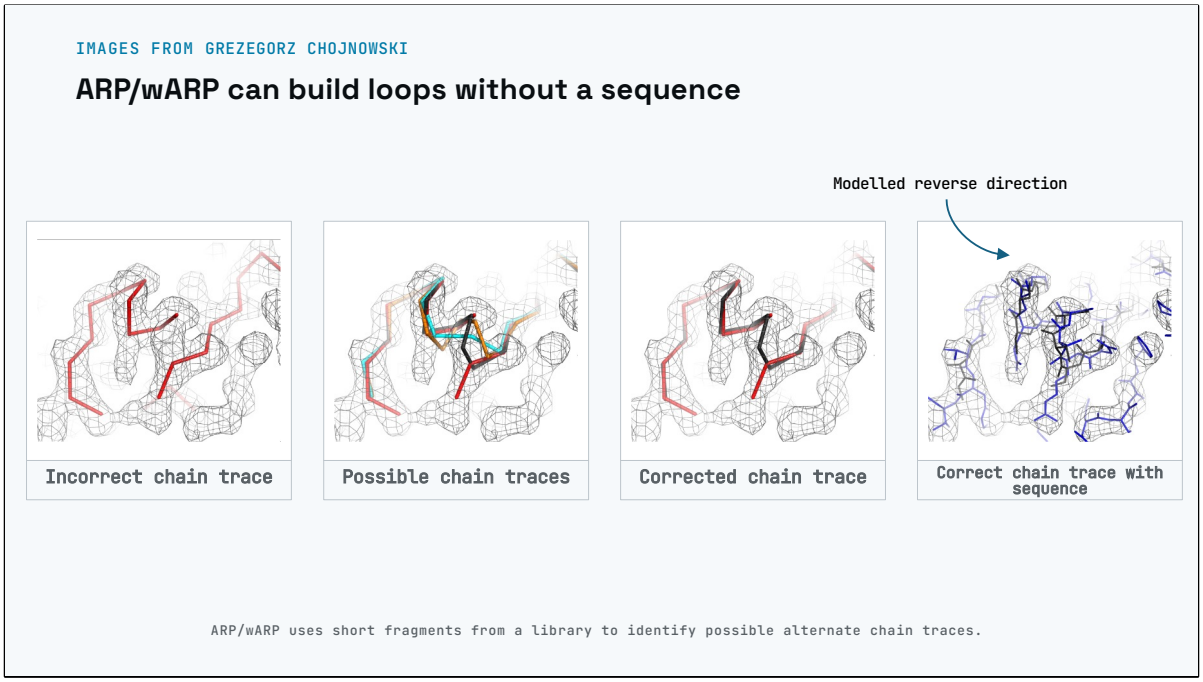
Appearance of density with varying contours



Distribution of volume for two amino acids

Modern ARP/wARP looks at the variation of density in sidechain areas to assign amino acid sequence.

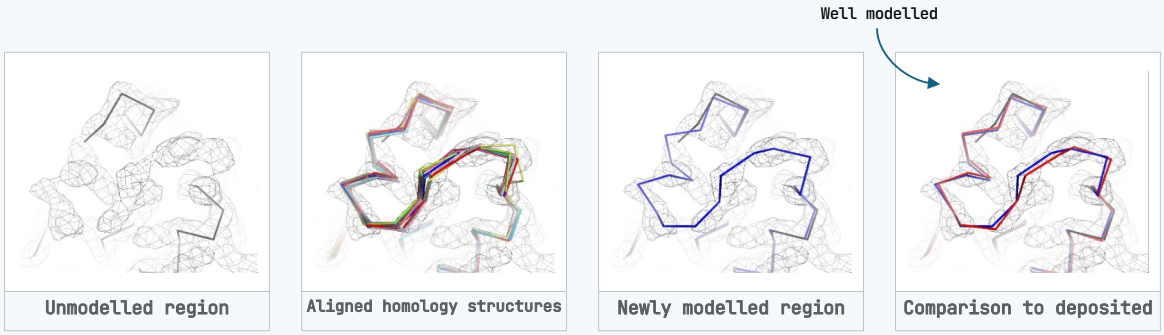
ARP/wARP used to decide the side chain type by looking at the patterns of free atoms in the density. However, this doesn't work very well at low resolution. The newer algorithm looks at the change in the density volume at different density thresholds (Chojnowski et al. 2019). In the chart we can see that the volume profiles of tryptophan and valine are quite different. The algorithm looks at the volume profile for the side unknown side chain and converts this into a probability for each side chain type.



ARP/wARP has a sequence-free loop building algorithm. Short fragments from a purpose-built library are superposed over the intermediate models. This algorithm is able to correct fragments that were built in the reverse direction (Chojnowski et al. 2019).

IMAGES FROM GREZEGORZ CHOJNOWSKI

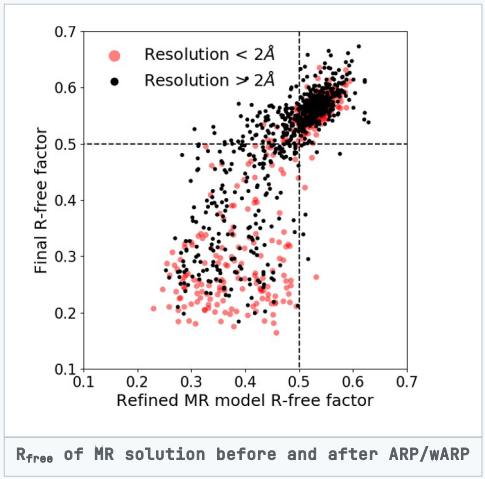
ARP/wARP can use homology to model missing regions



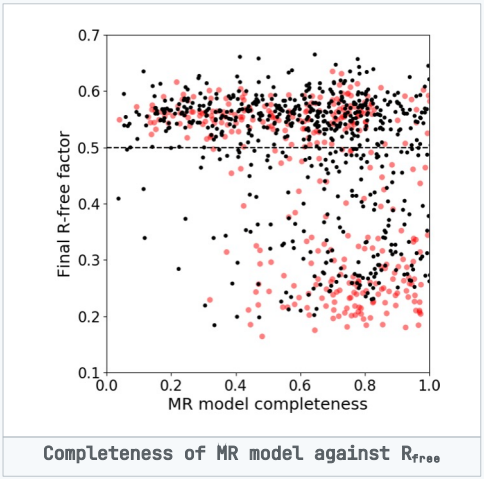
ARP/wARP superimposes homologous structures using GESAMT to help complete the model.

IMAGES FROM GREZEGORZ CHOJNOWSKI

ARP/wARP works well with good data



R_{free} of MR solution before and after ARP/wARP



Completeness of MR model against R_{free}

ARP/wARP works well with resolution better than 2 Å but can work lower with a good starting point.

This slide shows some results from ARP/wARP. The chart on the left compares the free-R factor of molecular replacement (MR) models before and after ARP/wARP. If the MR model has an R-free below 0.5 and the resolution is better than 2.0 Å then there is a good chance that ARP/wARP will be able to improve the model. If the resolution is worse 2.0 Å then a better initial MR model may be needed. The chart on the right shows the completeness of the initial MR model against the free-R factor after ARP/wARP. If the MR model has a very low completeness then the phases may be too poor for ARP/wARP to improve the model.

IMAGES FROM GREZEGORZ CHOJNOWSKI

When will ARP/wARP succeed, and how will I know?

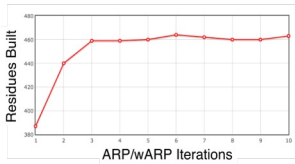
Things to check

Resolution is better than 2.6 A

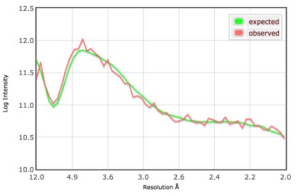
Space group is correct

Data completeness is high

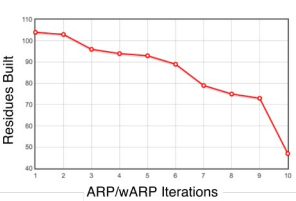
Data is **good quality**



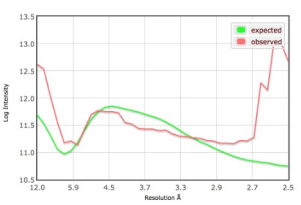
Successful ARP/wARP run



Normal Wilson plot



Unsuccessful ARP/wARP run



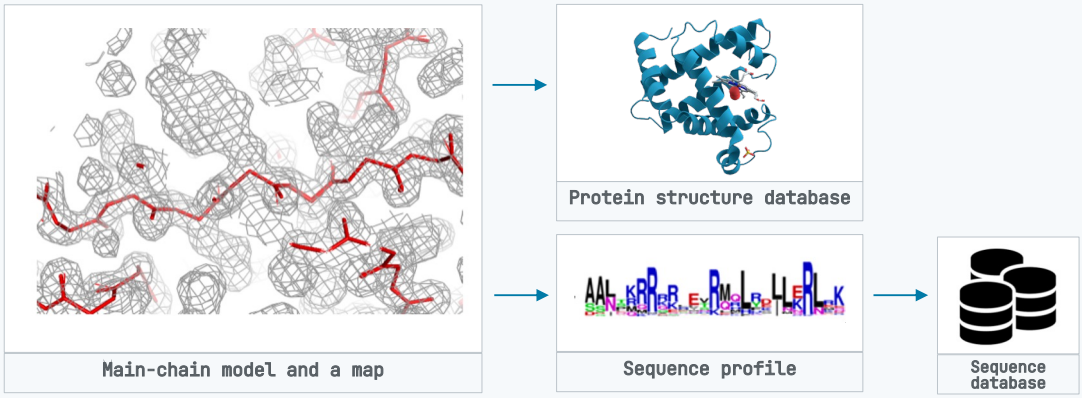
Abnormal Wilson plot

To enable a successful ARP/wARP run, ensure data is well processed.

Classic ARP/wARP works best when the resolution is better than 2.6 Å, but when the resolution is high it is able to build even if the initial phases are quite bad. The top left chart shows a good ARP/wARP run where the number of residues built goes up with the number of iterations. The bottom left chart shows what can happen if it does not work well. If you get a poor result but have high resolution data it a common error is that the space group is incorrect. It may be worth switching to a lower symmetry space group or trying different enantiomorphs if possible. ARP/wARP also can be sensitive to both the completeness and the quality of the data. The Wilson plot can indicate whether there are problems at specific resolutions. The top and bottom right charts show examples of good and bad Wilson plots.

IMAGES FROM GREZEGORZ CHOJNOWSKI

New developments: sequence identification from a map

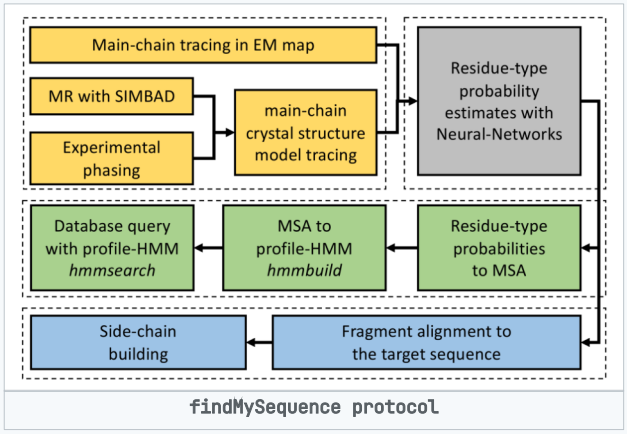


ARP/wARP can use the model and density to assign a sequence if it is not known.

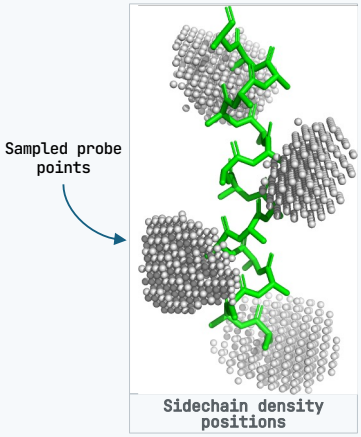
One useful thing to note is that ARP/wARP is able to build the backbone of a model without requiring a known sequence. There are two options to identify the protein from a backbone trace: you can use a program such as GESAMT, FATCAT or DALI to look for structural homologues using a 3D structure search or you can create a sequence profile from the density and search a sequence database. Grzegorz Chojnowski has written a program called findMySequence that does the latter.

IMAGES FROM GREZEGORZ CHOJNOWSKI

New developments: sequence identification with findMySequence



findMySequence protocol



findMySequence uses the density and sequence information to estimate a probability for each side chain.

If you have an EM map you can build the main chain without knowing the sequence of the structure. In crystallography, it is harder to solve the phase problem without knowing the sequence because we can't just use a homologue or a predicted model in molecular replacement. We can use SIMBAD for molecular replacement without a sequence or experimental phasing before building the main chain. Then findMySequence looks at a cloud of points in the density map where the side chain is expected to be and uses a neural network to turn these into probability estimates for each residue type. It then creates a hidden Markov model (HMM) to search sequence databases. Then the sequence can be aligned and the side chains built. There is also a program called checkMySequence that uses the same type probability estimates for validation and can be used to check for register errors.

IMAGES FROM GREZEGORZ CHOJNOWSKI

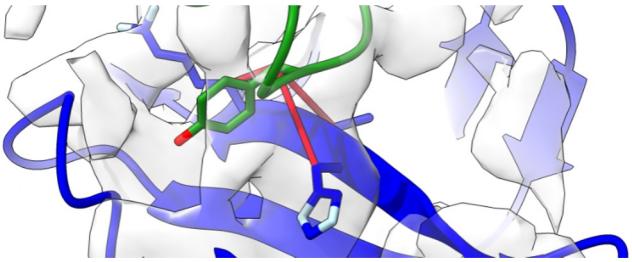
New developments: gapTrick

What is gapTrick?

AlphaFold3 often fails to predict weak or incomplete protein complexes

gapTrick uses multimeric templates to improve prediction reliability

Available in CCP4 Cloud

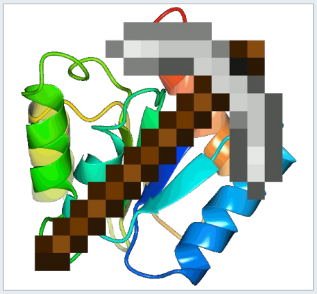


Iterative AlphaFold multimeric predictions with gapTrick can help correct errors

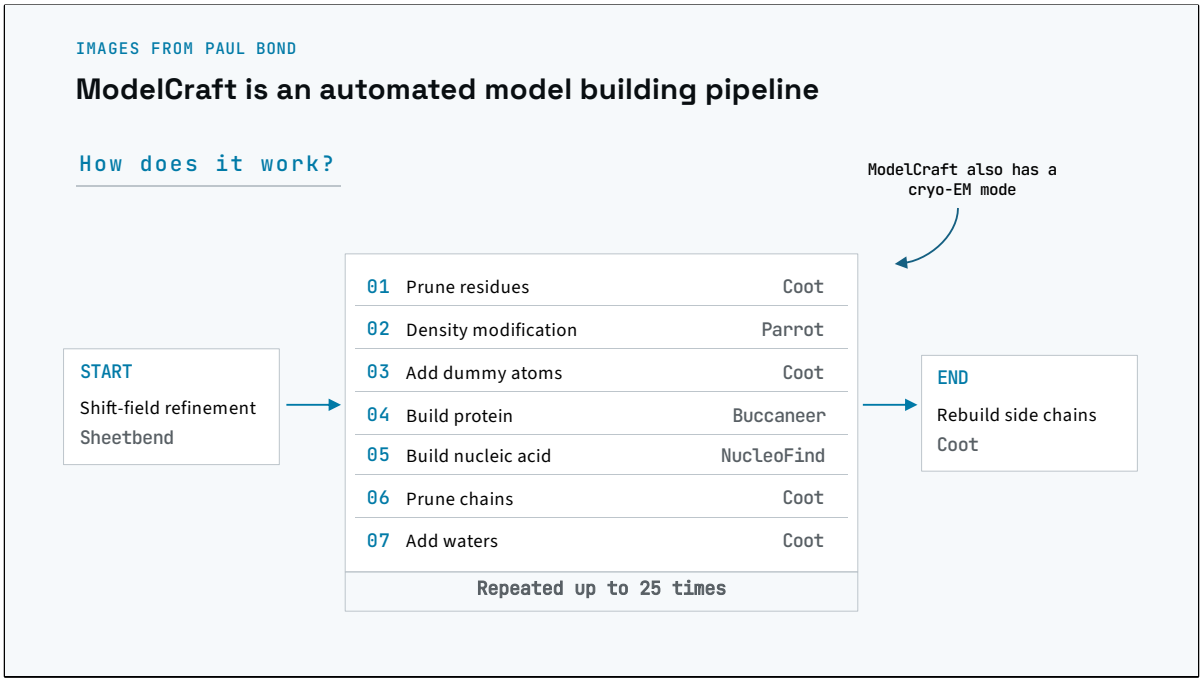
Another development from Grzegorz is gapTrick. AlphaFold models often have low confidence regions that need to be trimmed, and multi domain or multimeric complexes often need to be split up and placed separately due to errors in the relative positions of the domains/molecules. If you have models that have been placed or refined into a map then gapTrick runs AlphaFold2 using that as a template to improve the prediction, which is very helpful for building missing regions and correcting errors.

ModelCraft

Paul Bond - University of York
Kathryn Cowtan - University of York
Jordan Dialpuri - MRC LMB



Now we will look at ModelCraft, a pipeline that includes Buccaneer and NucleoFind along with others for density modification, refinement and validation.



ModelCraft was created to help build more structures when the initial phases are poor, such as after molecular replacement with a very incomplete or dissimilar model. It was also designed to build both protein and nucleic acids in the same pipeline. This is the overall ModelCraft pipeline as it stands. If you provide a starting model it is first refined using Sheetbend then Refmac. Then a single cycle of the pipeline has 7 steps, most of which are followed by refinement with Refmac. With high resolution data, it starts by pruning individual residues using Coot. Then it uses Parrot for density modification, which is especially powerful if there is non-crystallographic symmetry that can be determined from the current model. After Parrot, it adds dummy atoms using Coot to improve the phases, but the dummy atom structure is only accepted if R-free improves. The dummy atoms are removed from the model then Buccaneer builds the protein and NucleoFind builds DNA and RNA. Coot is used to remove incorrect chain fragments and to add waters, but again only accepting the result if R-free improves. Once the model stops improving (or it reaches 25 cycles) the cycle with the best model is used as the output. If the resolution is high then Coot is used to rebuild missing or incorrect side chains.

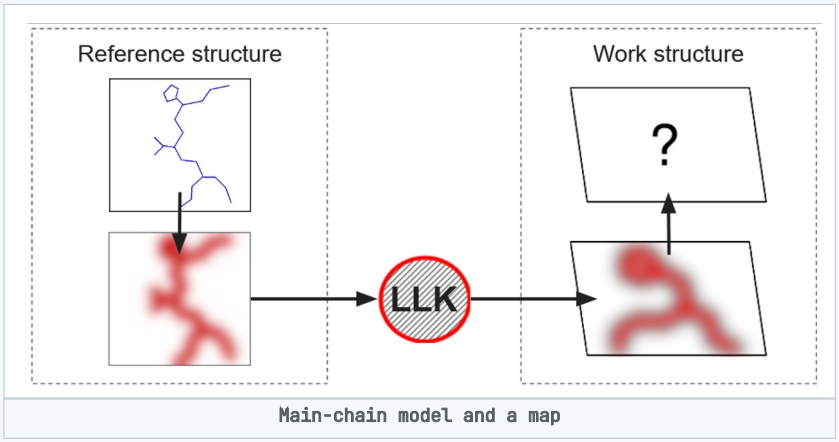
Buccaneer

Kathryn Cowtan - University of York

IMAGES FROM PAUL BOND

Buccaneer is a protein model building tool

How does it work?

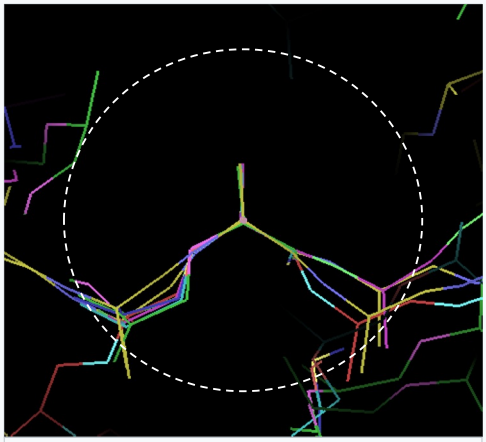


Main-chain model and a map

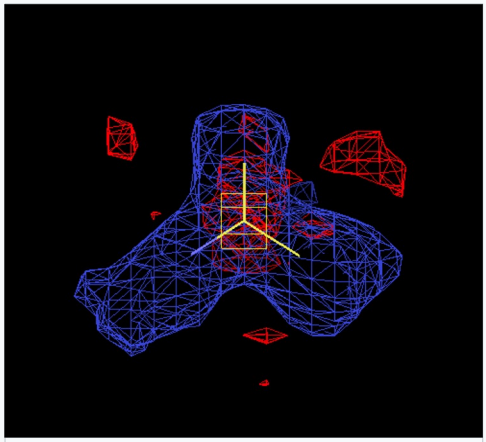
Buccaneer uses a known reference structure in order to help it build at both high and low resolution. You give Buccaneer the map you're trying to build a model into (the work map) and it generates a map for a reference structure with the same resolution and same level of noise as this map. Then it uses the reference structure to find out what features in the reference map look like by creating a log-likelihood (LLK) target. The LLK target is then used to search for the same features in the work map.

IMAGES FROM PAUL BOND

Buccaneer is a protein model building tool



4 A sphere target function



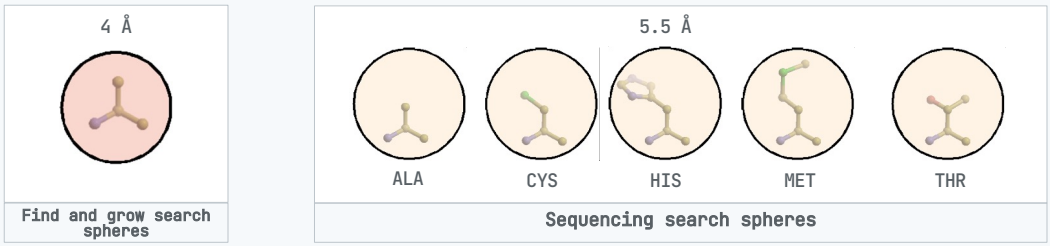
High (blue) and low (red) conserved regions

Buccaneer uses a target function to search for potential C alpha positions in the density.

The target to search for individual residues is obtained by superimposing all the residues in the reference structure on top of each other. The mean and variance of the density is then used to construct a likelihood target within a 4 Å sphere. The blue density shows regions of conserved high density and the red density shows regions of conserved low density. Both are important when finding residues in the work map. If there is low density in the blue regions or high density in the red regions then it is likely not the correct position/orientation for a residue.

IMAGES FROM PAUL BOND

Buccaneer is a protein model building tool



Buccaneer uses a 4 Å target for C alphas and 5 Å target for side chains.

The 4 Å C-alpha target is used in the first step of the program to find initial residue positions and the second step to grow those residues into chains. The program also uses 5.5 Å targets centred around C-beta atoms. A different target is made for each residue type and those are used for sequencing. Rotamers are not separated before making the targets, so the mean and variance of the density is calculated for side chains pointing in lots of different directions, but this still provides enough information to find the correct sequence.

IMAGES FROM PAUL BOND

Buccaneer is a protein model building tool

How does it work?

01	**Find** oriented residue positions
02	**Grow** each residue into a chain fragment
03	**Join** overlapping fragments and resolve branches
04	**Link** nearby termini
05	**Sequence** the chains
06	**Correct** insertions and deletions
07	**Filter** to remove short chains
08	**NCS** superposition to extend chains
09	**Prune** residues to resolve clashes
10	**Rebuild** side chains

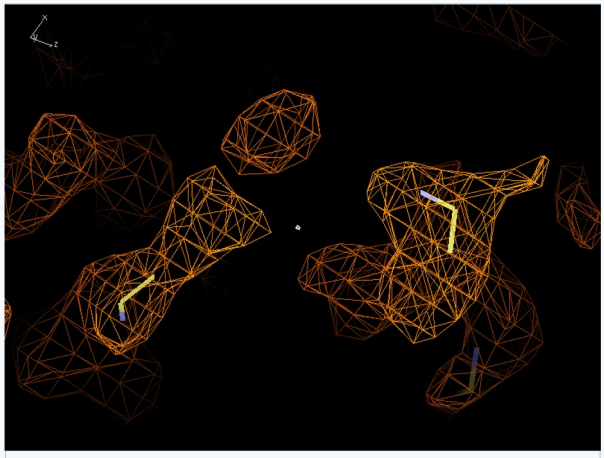
Buccaneer and *Nautilus* both run as a cyclic calculation with multiple internal steps. These are the 10 steps in *Buccaneer*. The steps for *Nautilus* are similar.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	Find
02	Grow
03	Join
04	Link
05	Sequence
06	Correct
07	Filter
08	NCS
09	Prune
10	Rebuild

Buccaneer steps



R-free: 54 %

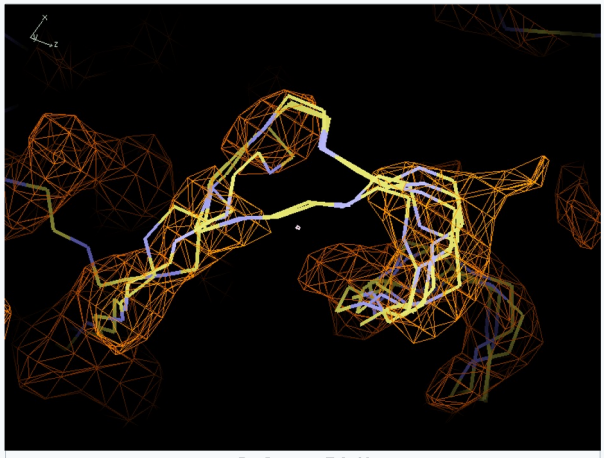
This is an example of *Buccaneer* building a difficult loop in a 2.9Å map. The first step is to find possible positions for new residues. It has placed residues at three positions in this figure (two in the foreground and one in the background) where the density has the right shape according to the C-alpha target function. There are also other residues placed out of this view.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	Find
02	Grow
03	Join
04	Link
05	Sequence
06	Correct
07	Filter
08	NCS
09	Prune
10	Rebuild

Buccaneer steps



R-free: 54 %

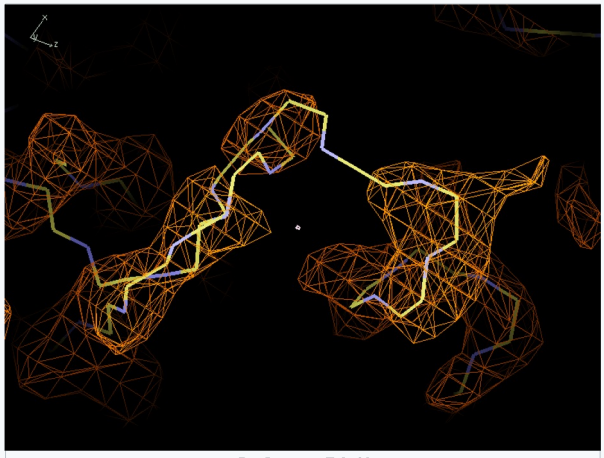
The next step is to grow each residue into a long chain fragment. New residues get added at both ends using an exhaustive search over allowed Ramachandran angles and the same C-alpha target. Once the target score drops below a threshold value the growing stops. This step gives lots of fragments, many of which overlap. In this example it has built a helix on the right where the chains agree fairly well, two different paths to bridge the gap in the middle and a contradictory chain on the left that has been built in the opposite direction.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

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09	Prune
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Buccaneer steps



R-free: 54 %

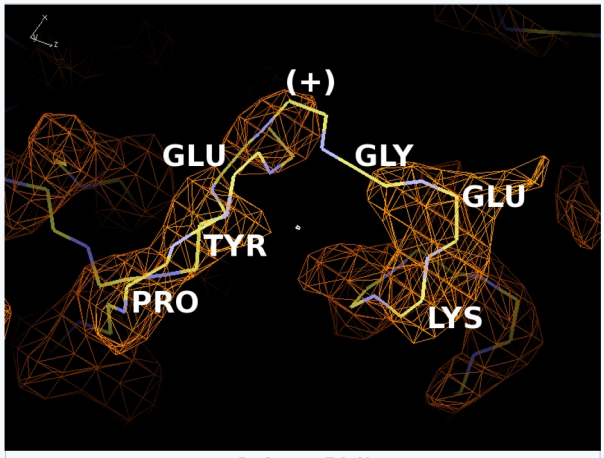
The joining step merges all the chain fragments that agree with each other into single chains. This step also has to make decisions about which routes to follow. It tries to build the longest chains (e.g. it chose the longest route over the middle loop), which helps to build helices correctly. The chain built in the reversed direction is still there because it can't be merged and at this stage it hasn't been decided which is correct.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

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Buccaneer steps



R-free: 54 %

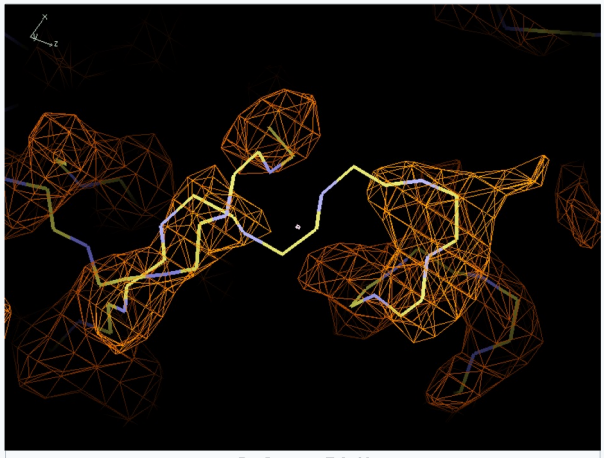
We have skipped the linking step because there are no termini that can be linked in this view. The sequencing step works by assigning each residue a probability that it is each of the 20 residue types using the C-beta targets. These probabilities are then compared to the known sequence(s) to look for continuous runs that stand out with a high probability. In this example, the sequence only fits the chain well if an extra symbol is added in the middle. This is called an insertion and (unless the sequence is wrong) it means the chain has been built incorrectly. The reversed chain on the left wasn't sequenced as there was no part of the sequence that fits well.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	Find
02	Grow
03	Join
04	Link
05	Sequence
06	Correct
07	Filter
08	NCS
09	Prune
10	Rebuild

Buccaneer steps



R-free: 54 %

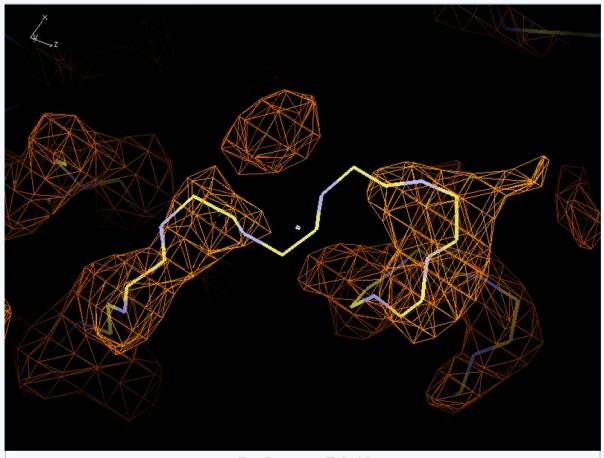
The next step is to correct any insertions (where it has built one extra residue) and deletions (where it has built one fewer residue) found during the sequencing step. The insertion that was over this loop as been corrected by removing three residues and rebuilding two.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	Find
02	Grow
03	Join
04	Link
05	Sequence
06	Correct
07	Filter
08	NCS
09	Prune
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Buccaneer steps



R-free: 54 %

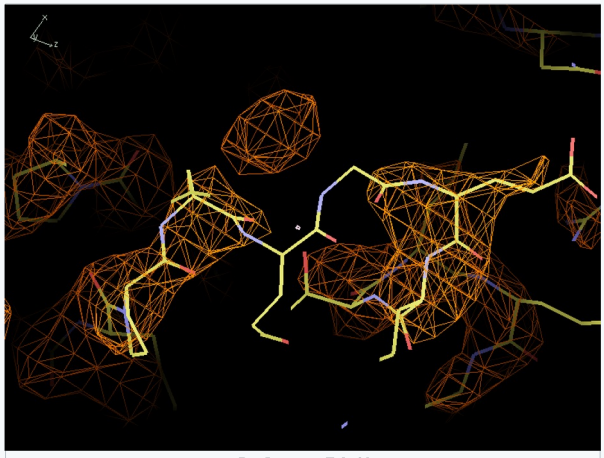
The pruning step removes residues from clashing chains to produce a single consistent model. If the pruned chains have less than 6 residues remaining they are removed too. Pruning is done at this late stage to have the most information about which chain is correct. Residues that are unsequenced or are from shorter chains with worse fit to the density will be removed preferentially. In this case the reversed chain on the left was removed.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	Find
02	Grow
03	Join
04	Link
05	Sequence
06	Correct
07	Filter
08	NCS
09	Prune
10	Rebuild

Buccaneer steps



R-free: 54 %

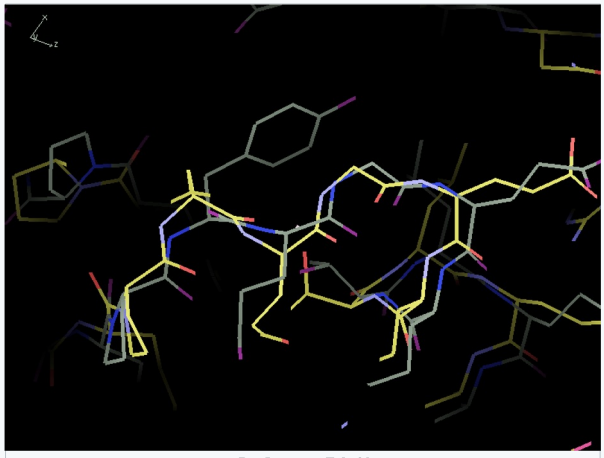
Finally, the last step is to build side chains for each residue using a rotamer library. The rotamer with the highest average density at the atomic positions is chosen. If two side chains end up clashing, *Buccaneer* will try and find a pair of rotamers that do not clash. If this fails both residues will be truncated to C-beta.

IMAGES FROM KATHRYN COWTAN

An example of Buccaneer in action

01	**Find**
02	**Grow**
03	**Join**
04	**Link**
05	**Sequence**
06	**Correct**
07	**Filter**
08	**NCS**
09	**Prune**
10	**Rebuild**

Buccaneer steps

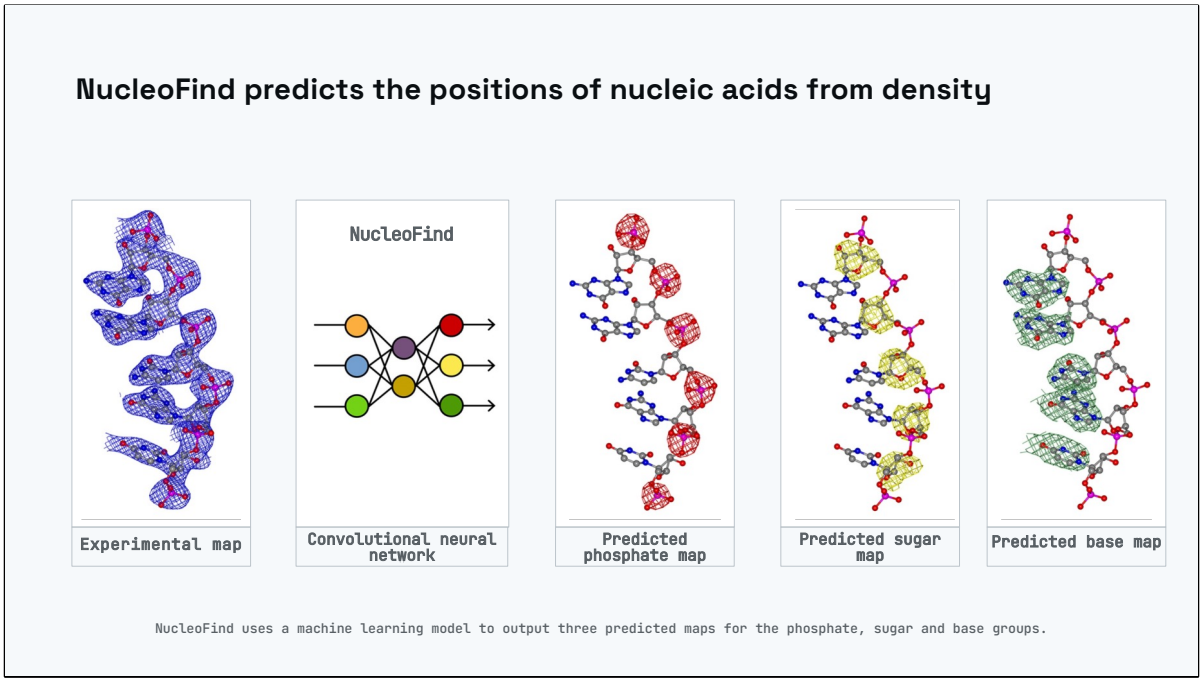


R-free: 54 %

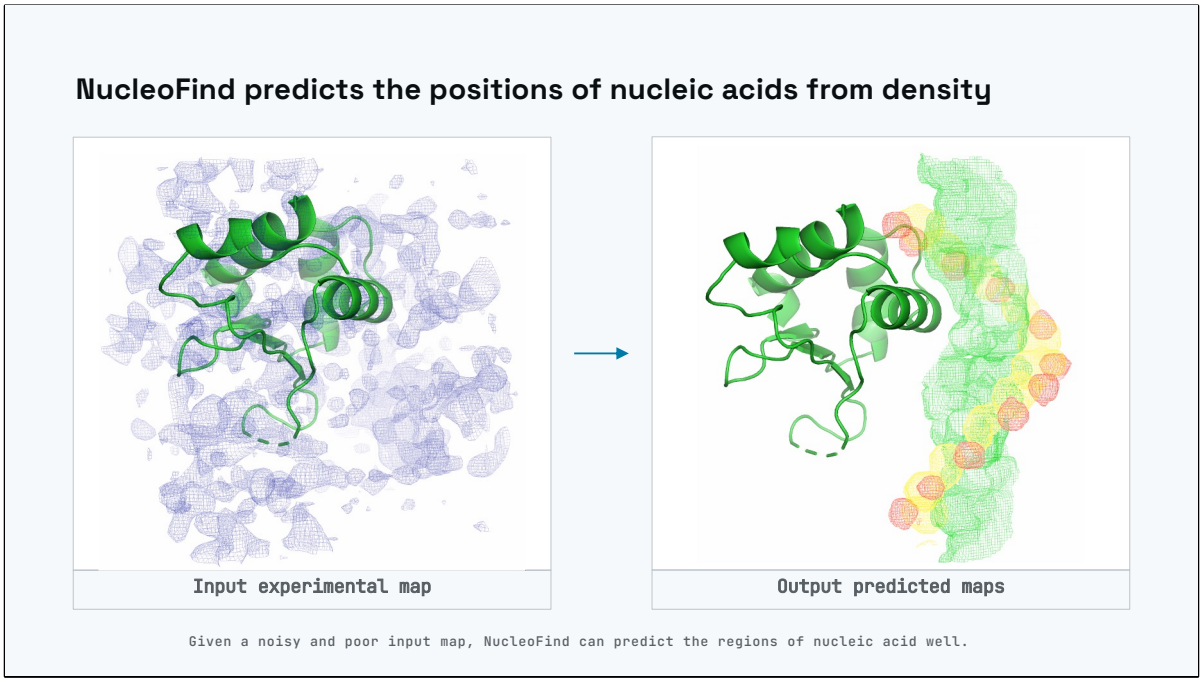
This shows a comparison with the deposited model. *Buccaneer* built most of the model quite well but got some things wrong, for example the the tyrosine side chain. The model will improve after refinement with *Refmac* and further cycles of *Buccaneer*.



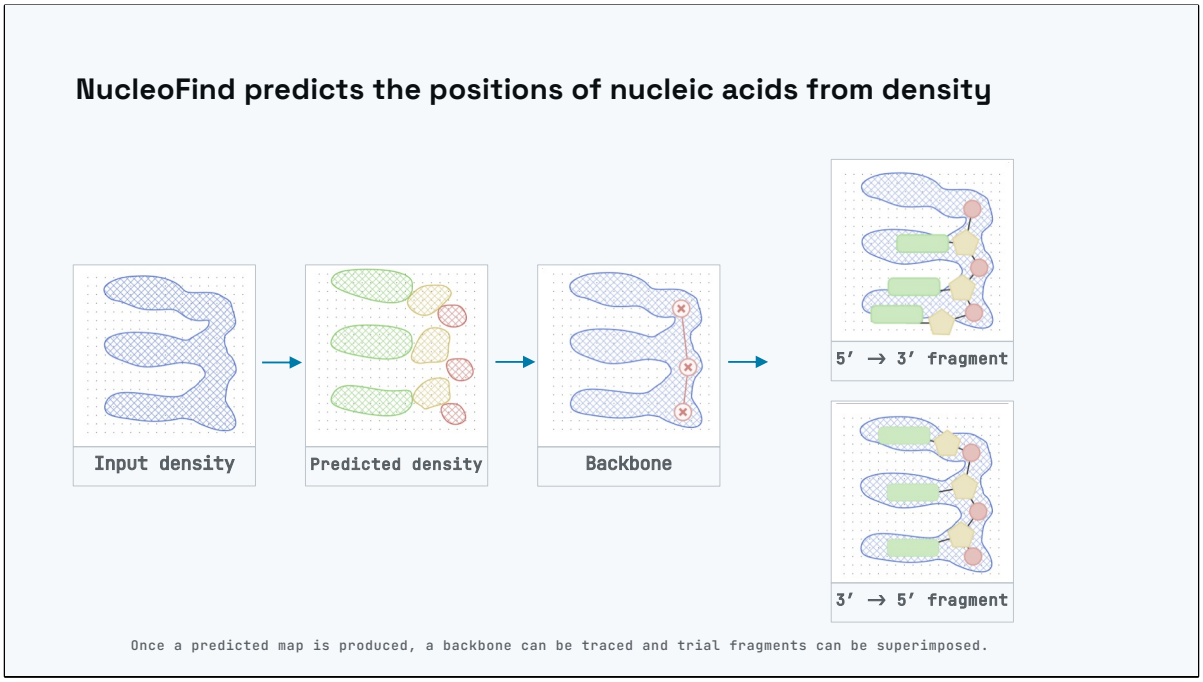
Buccaneer is the program ModelCraft uses for protein building. NucleoFind (Jordan Dialpuri) is the program it uses for building nucleic acids. NucleoFind uses a deep-learning algorithm for locating phosphate, sugars and bases in maps. It is a multi-class U-Net to produce three separate maps, one for each type of feature, in a single prediction.



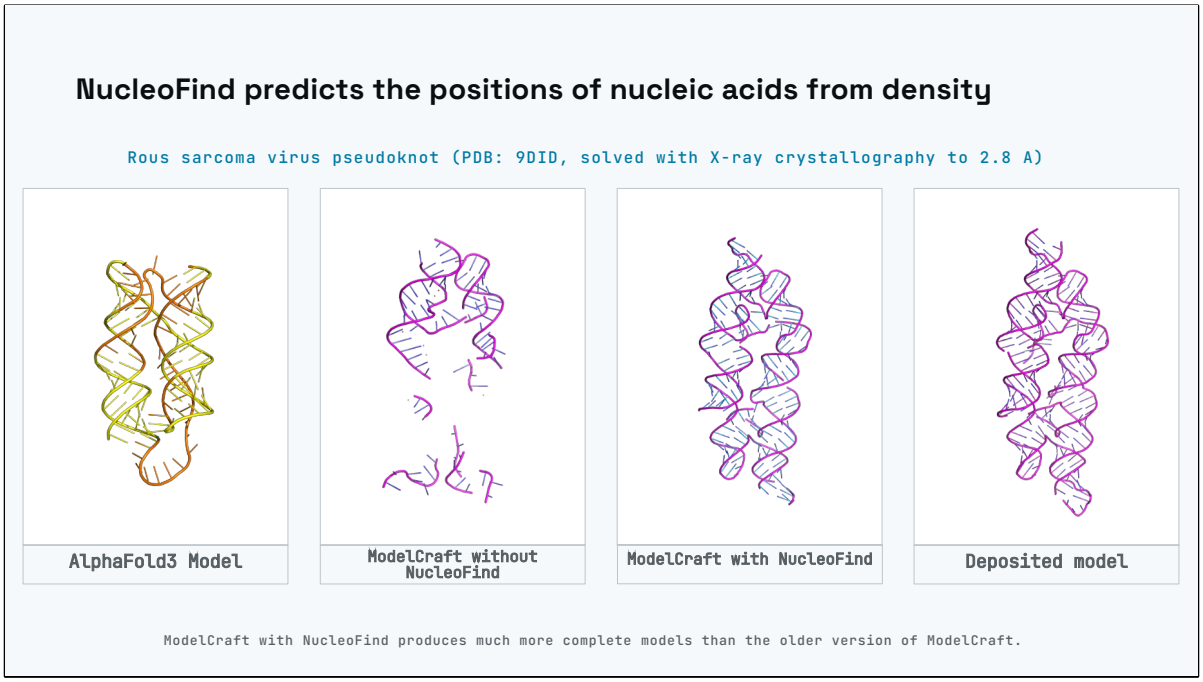
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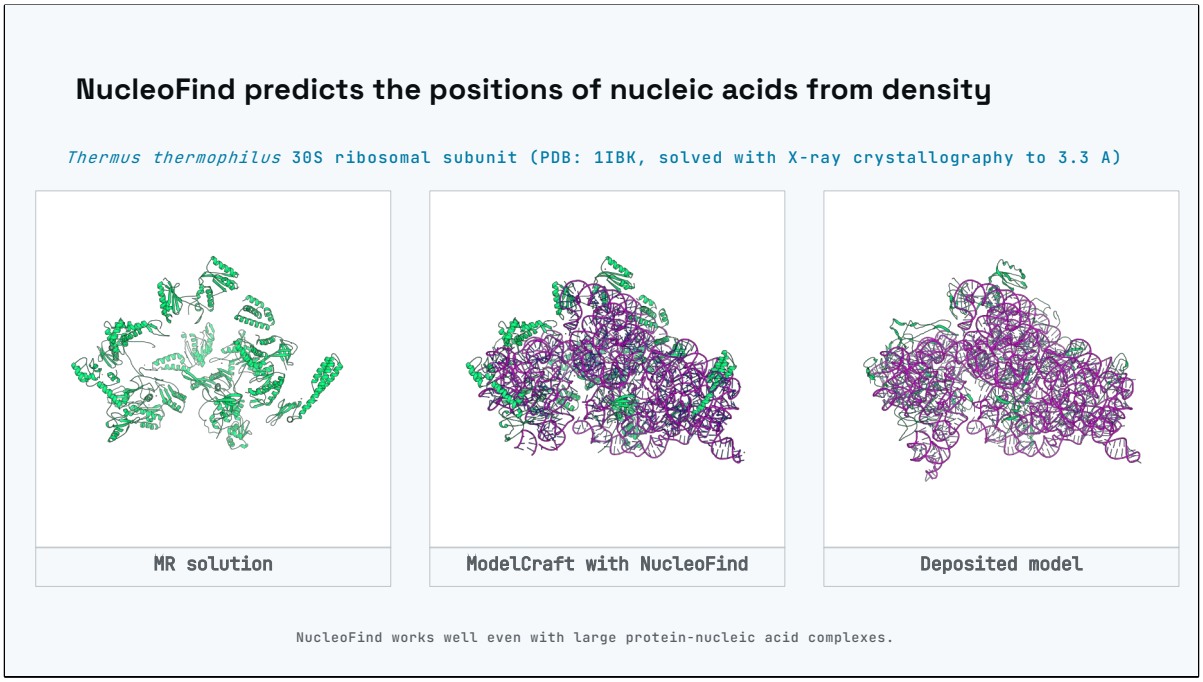
NucleoFind can work on maps that are not easy to interpret by eye. This examples is a protein-DNA complex at 2.4 Å resolution. It was phased by refining only the protein so the density for the DNA is not clear. NucleoFind still predicts the features of the DNA well.



NucleoFind uses these predicted density maps to help build atomic models of nucleic acids. This gives an overview of the procedure. The backbone is located using the phosphate predictions, then a library of tri-nucleotide fragments is superposed in both the 5’ and 3’ directions to chose the best fragment for each direction. Overlapping tri-nucleotides are combined into longer chains and the best direction is chosen. There are further steps for growing chains, sequencing and adding bases, which are currently the same as they were in Nautilus, which was the previous program used for nucleic acid building in ModelCraft.



ModelCraft now uses NucleoFind for building nucleic acids instead of Nautilus. This shows a comparison of ModelCraft with Nautilus and ModelCraft with NucleoFind on a 2.8 Å X-ray RNA structure. The AlphaFold model is not very accurate and the structure was solved by experimental phasing using iridium atoms. ModelCraft with Nautilus produced a model which is 31% complete, but the model from ModelCraft with NucleoFind was 86% complete, with a run time of only 30 minutes.



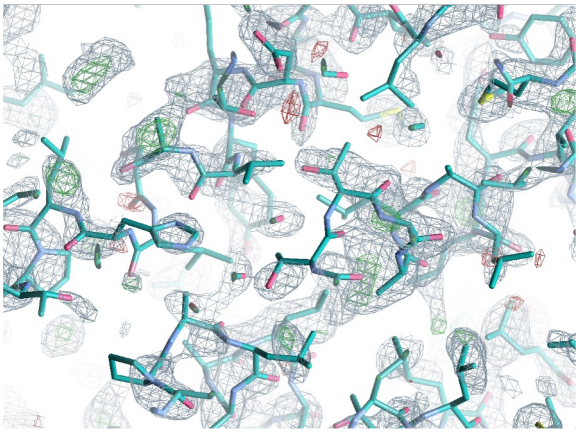
This example is a large 3.3 Å X-ray structure of a protein, nucleic acid complex. The MR solution with the protein only did not give great phases because of the missing RNA. Nautilus only managed to build a few residues, but ModelCraft with NucleoFind built a model that was 88% complete after 25 cycles.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 0

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 54 %

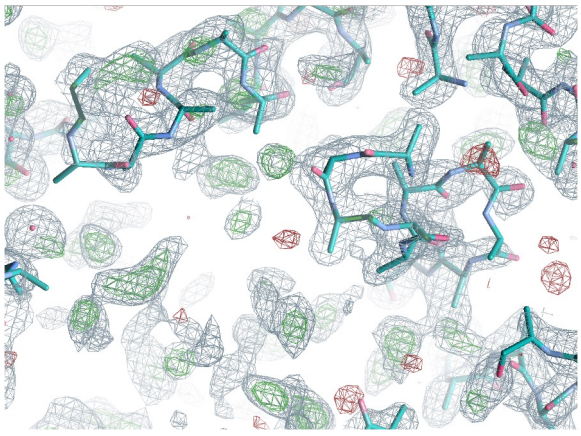
This is a difficult example where *ModelCraft* worked well. The target structure is 3GVP with 4 copies of a 435 residue protein and data to 2.25 Å resolution. The *AlphaFold* model was trimmed to remove residues with high pLDDT but it was not split into separate domains for molecular replacement. *MOLREP* placed four copies but only three of them were placed correctly. The F-map correlation after MR was 0.3, which is on the threshold of what would be considered a solution. The phases are made worse by the relative domain positions being slightly different in the true structure and the *AlphaFold* model. This is a view over the incorrectly placed copy after refinement with *Sheetbend* and *Refmac*, with the R-free value stuck around 54%.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 3

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 50 %

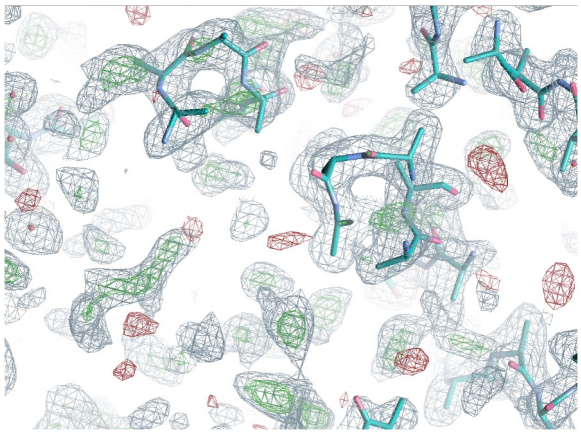
This is the same view but after 3 cycles of *ModelCraft*. As you can see, the model and map look completely different. The original structure in this region has been removed and now there are just a few small fragments built by *Buccaneer*. R-free has reduced slightly from 54% to 50%.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 4

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 50 %

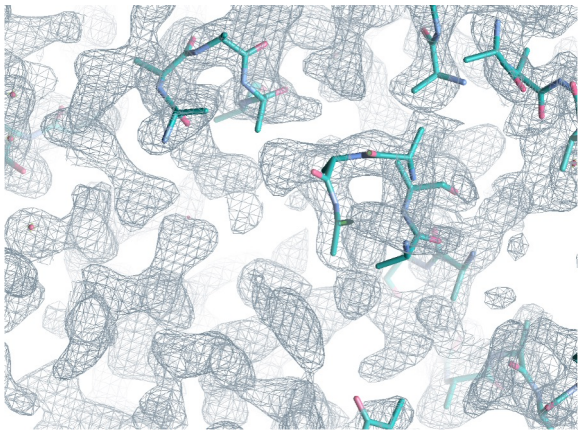
A large improvement is seen in the 4th cycle. The first step is to remove the worst residues using *Coot* and refine again. This step is only done if the resolution is better than 2.3 Å. The density for the removed residues in the top left disappeared after further refinement, suggesting that the density was biased.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 4

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 50 %

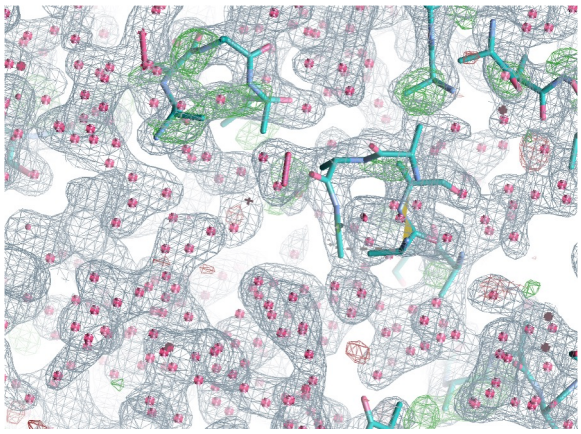
The second step is density modification using *Parrot*, which is especially powerful when there are multiple copies of a chain and the non-crystallographic symmetry (NCS) operators can be determined from the current model. NCS-related regions of the map will be averaged, weighted by how similar the density is. The area that gets averaged tends to grow as the phases improve. In this case, *Parrot* only determined symmetry operators between the three correctly-placed copies and not the copy in this region, but there is still a very big improvement to the density.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 4

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 32 %

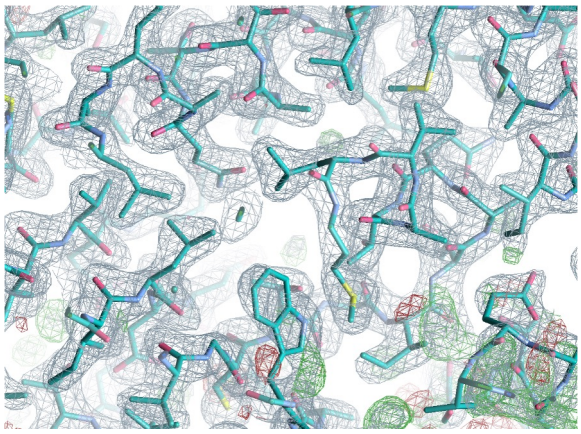
The third step adds dummy atoms into the *Parrot* map using *Coot*. The existing structure and the dummy atoms are then refined with *Refmac*. No attempt is made to interpret the dummy atoms as it is a difficult task and at lower resolutions it is unlikely that they correspond to atom positions. Instead, they only are used as a form of density modification. The resulting map is only accepted if R-free improves from the previous refinement of the protein model, which it did in this case from 50% to 32%.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 4

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot



R-free: 33 %

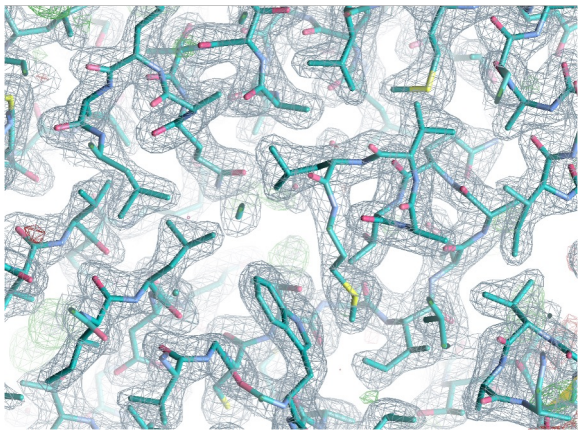
The dummy atoms are then discarded and *Buccaneer* is used to build a protein model. Because of the better phases, it has built most of this view correctly apart from some parts in the bottom right. R-free has increased slightly (from 32% to 33%) compared to the dummy atom model, but there will be much less overfitting as it is now a protein model with restraints applied.

IMAGES FROM PAUL BOND

An example of ModelCraft in action

Cycle 5

01	Prune residues	Coot
02	Density modification	Parrot
03	Add dummy atoms	Coot
04	Build protein	Buccaneer
05	Build nucleic acid	NucleoFind
06	Prune chains	Coot
07	Add waters	Coot

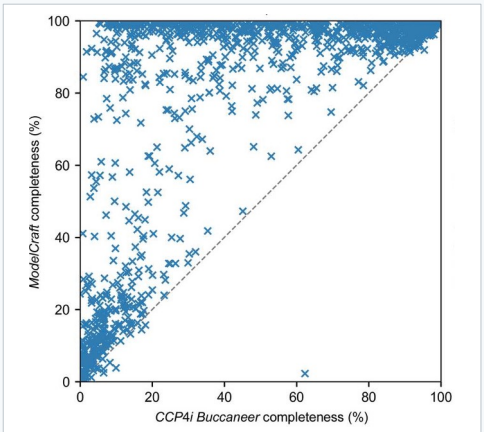


R-free: 26 %

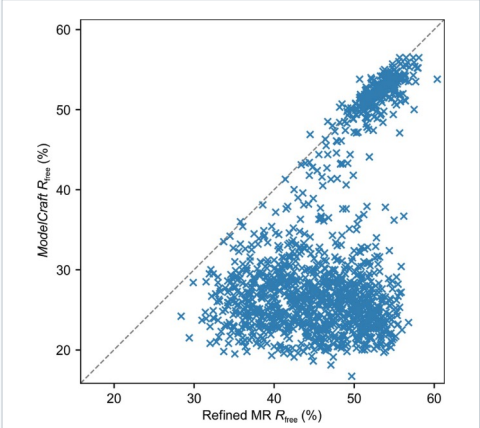
At the end of cycle 5 most of those issues have been fixed and R-free has improved to 26%. In this case, this is the best structure that ModelCraft produces. It still requires some interactive rebuilding and validation using in *Coot*. For example, there is a twisted peptide bond in the bottom right as indicated by the yellow triangle.

IMAGES FROM PAUL BOND

Results of ModelCraft



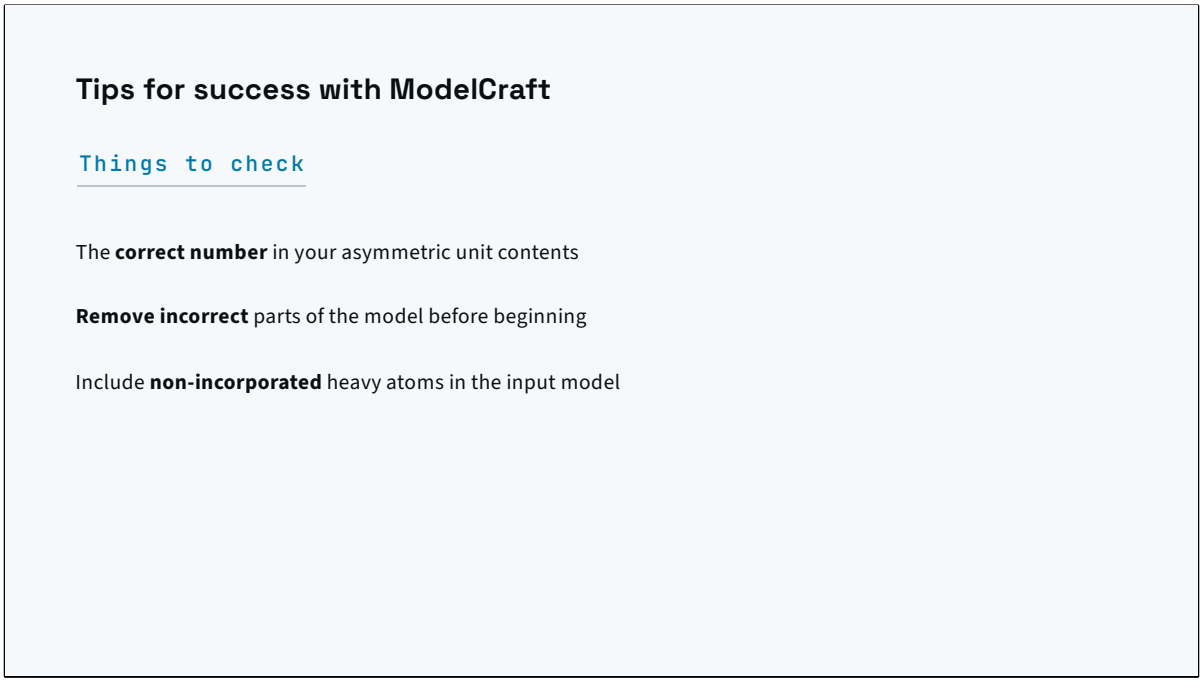
Completeness versus older pipeline



R_{free} versus older pipeline

ModelCraft produces much more complete models than older methods.

These charts show the performance of ModelCraft on a large molecular replacement test set with resolutions spread evenly between 1 and 3.5 Å and examples with both good and bad phases. On the left is a comparison of the completeness of protein models after the old Buccaneer pipeline from CCP4i and ModelCraft. There are a lot of cases at the top of the chart where ModelCraft produces a much more complete model. There is one case in the bottom of the chart where the old pipeline produces a much more complete model, but this issue was fixed in a later version of ModelCraft. On the right is a comparison of the free-R factor before and after ModelCraft. ModelCraft is sometimes able to build a good model even if R-free after MR is as high as 55%. If the MR model has an R-free below 50% then there is a good chance that ModelCraft will be able to improve it.



When running ModelCraft, it is important to have the correct number of copies in order to calculate the solvent content for Parrot, otherwise density modification may flatten weak protein regions or amplify noise in the bulk solvent. If you know that part of your model is wrong, then it is probably better to remove the incorrect residues first. It can also be useful to include non-incorporated heavy atoms (e.g. not Se from MSE) to improve the phases and to stop the model being built into that density.

SHELXE

George Sheldrick (1942 – 2025) – University of Gottingen
Isabel Usón – IBMB-CSIC

The next program we will look at is SHELXE.

IMAGES FROM USÓN AND SHELDRICK (2022)

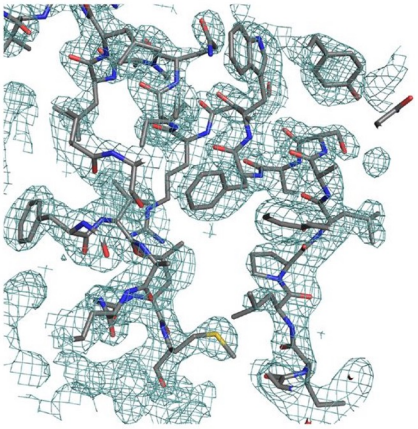
SHELXE is capable of building complete protein models

Origins of SHELXE

Originally made for **density modification** (2002)

Main-chain tracing was added later (2010)

Recently **side-chain tracing** was added (2020)



Automatically modelled protein from SHELXE

SHELXE was originally a density modification program that uses a sphere-of-influence algorithm that works better at high resolution, similar to density modification with ACORN. In 2010 a main-chain tracing algorithm was added (as building an atomic model is a good way to improve phases) and recently (in 2020) side-chain tracing algorithms were also added, so that now SHELXE is capable of building complete protein models.

IMAGES FROM USÓN AND SHELDRICK (2022)

SHELXE is capable of building complete protein models

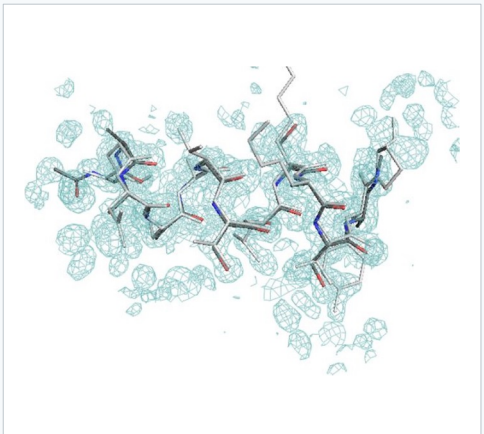
How does it work?

01	Create a ‘no-go’ map from symmetry elements and model
02	Search for alpha-helices and common fragments
03	Extend by looking two residues head with common ϕ/ ψ pairs
04	Decide whether to accept each chain
05	Combine overlapping chains through cutting and splicing
06	Bridge gaps in the trace
07	Refine per-residue B-factors

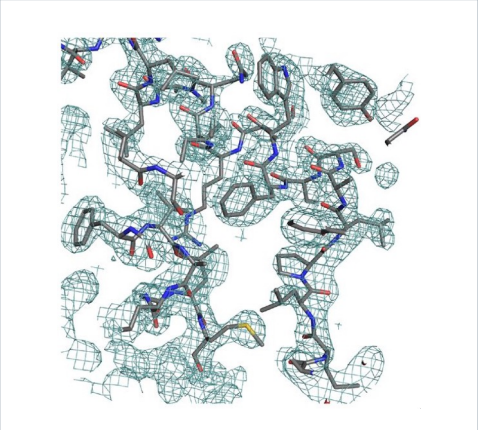
Main-chain tracing in SHELXE works by first creating a “no-go” map where the algorithm is not allowed to build into. Initially, this only includes symmetry elements and any starting model provided, but as new chains are grown and accepted this area expands. Alpha helices and common fragments are placed into unmodelled regions of density and minimised using a weighted density score. Fragments are then extended using a two-residue look-ahead approach that minimises common ϕ/ψ pairs using the same weighted density score. Extended chains are accepted based on their density score, length, Ramachandran outliers, secondary structure, and possible hydrogen bonding from N atoms. Overlapping chains are combined by cutting both chains at the point that they are closest and joining the best fragments from either side. There is also a step to bridge small gaps between close fragments. After building, per-residue B-factors are refined and a new map is produced for further building cycles.

IMAGES FROM USÓN AND SHELDRICK (2022)

SHELXE can trace side chains



Gamma extension



Full sequencing

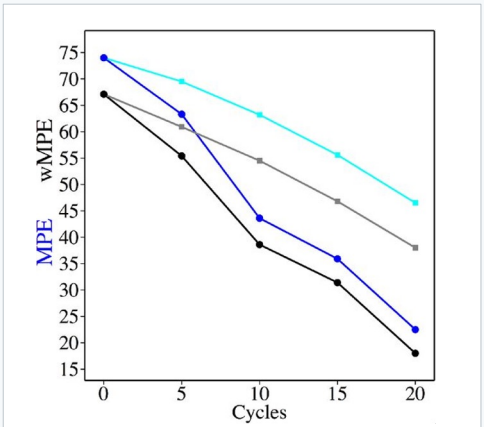
SHELXE can work with and without the sequence to model side chains.

Two side-chain building algorithms were added to SHELXE: one that does not require a sequence, which works by looking at the density for gamma atoms if the χ1 angle was +60º, -60º and 180º; and also a sequencing algorithm that aligns the built main chain to the sequence and builds the full side chains.

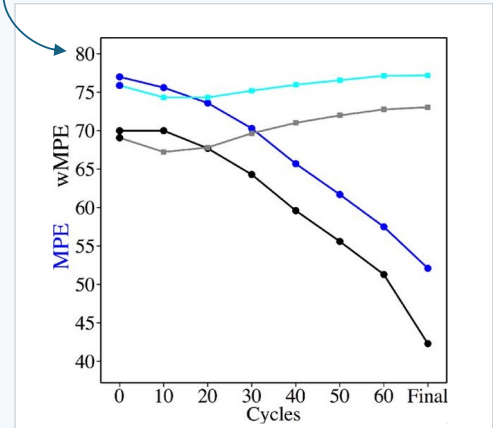
IMAGES FROM USÓN AND SHELDRICK (2022)

Results of SHELXE

Autotracing helps reduce MPE in this example



DNA-binding protein at 0.95 A



Human myosin 5b at 2.07 A

In certain cases, auto tracing can meaningfully improve the mean phase error.

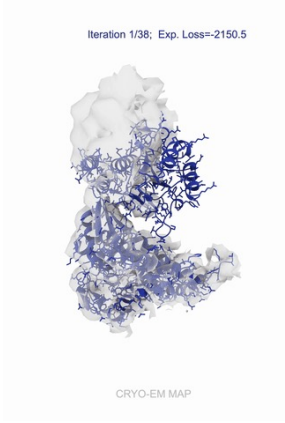
These graphs show the mean phase error for two examples with (dark lines with circles) and without (light lines with squares) autotracing. The example on the left is 1ZZK, a DNA-binding protein with 0.95 Å resolution data. It was phased using only four residues. In this case, low phase errors can be achieved with and without autotracing (after more cycles). The example on the right is 4J5M, human myosin at 2.07 Å resolution. Here SHELXE can only improve the phases if autobuilding is used. Changes to the main-chain building algorithms allow it to work at lower resolutions: typically better than 2.5 Å for molecular replacement and better than 3 Å for experimental phasing (Usón and Sheldrick, 2018). At higher resolutions it is very useful for improving poor phases.

What else can you use?

Other automated model building programs

Phenix

Phenix suite contains many tools



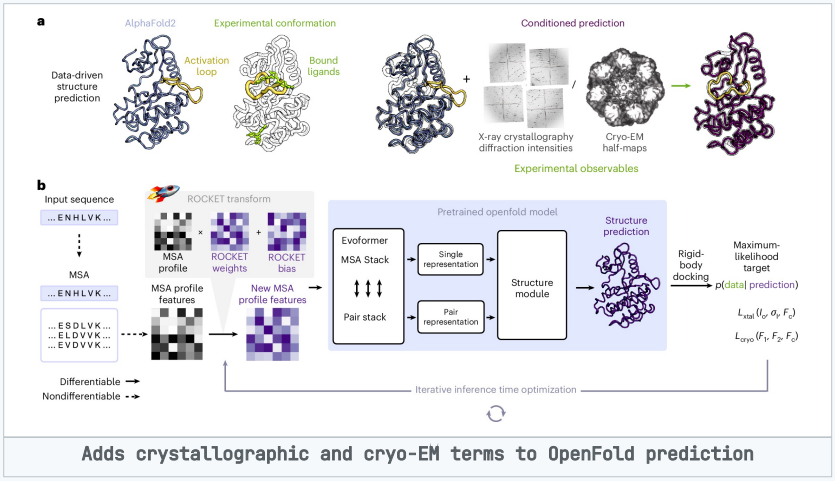
New development – ROCKET

Phenix and ROCKET are two of many alternate model building programs available for use.

You could use tools in the Phenix suite – Phenix AutoBuild, Phenix PredictAndBuild both may work, or you make morph a predicted model with a new program like ROCKET.

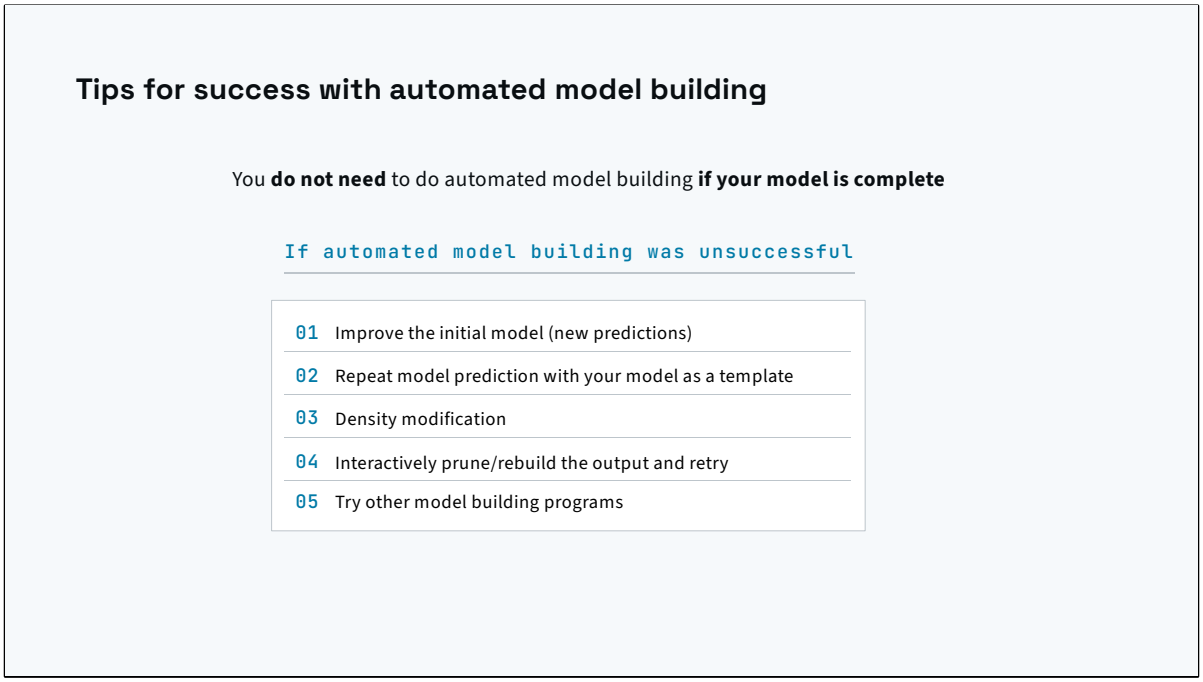
IMAGES FROM FADINI (2026)

ROCKET: Data-driven structure prediction refinement with ROCKET



Adds crystallographic and cryo-EM terms to OpenFold prediction

ROCKET iteratively reruns OpenFold predictions while considering the available experimental data, which often leads to morphing of the predicted structures to fit the experimental data (density map).



If the program you use hasn't done a very good job at building there are a number of things to try. You can try other model prediction programs or try to process the current prediction differently, e.g. with different levels of trimming or splitting up domains. As shown with Phenix PredictAndBuild and gapTrick, repeating structure prediction with a refined/rebuilt template can improve the prediction. The starting phases might be improved through density modification. In EM you could try post-processing methods to improve the map. You might also be able to do some interactive model building to improve either the input model or the partially-built output model. Other model-building programs could be used with the same input data or with the partially-built model.

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